

The Program in Action

When you run *dicemath.py*, the output will look like this:

Dice Math, by Al Sweigart al@inventwithpython.com

Add up the sides of all the dice displayed on the screen. You have 30 seconds to answer as many as possible. You get 4 points for each correct answer and lose 1 point for each incorrect answer.

Press Enter to begin...

```
+-----+
| 0  0 |
|  0   |
| 0  0 |
+-----+
```

```
+-----+
|      0 |
|      |
| 0      |
+-----+
```

```
+-----+
| 0  0 | +-----+
|  0   | | 0      |
| 0  0 | +-----+
+-----+ |      0 |
+-----+
```

Enter the sum: **13**

--snip--

How It Works

The dice on the screen are represented by a dictionary stored in the `canvas` variable. In Python, tuples are similar to lists, but their contents cannot be changed. The keys to this dictionary are (x, y) tuples marking the position of a die's top left corner, while the values are one of the “dice tuples” in `ALL_DICE`. You can see in lines 28 to 80 that each dice tuple contains a list of strings, which graphically represents one possible die face, and an integer of how many pips are on the die face. The program uses this information to display the dice and calculate their sum total.

Lines 174 to 177 render the data in the `canvas` dictionary on the screen in a manner similar to how Project 13, “Conway’s Game of Life,” renders cells on the screen.

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1. `"""Dice Math, by Al Sweigart al@inventwithpython.com`
 2. `A flash card addition game where you sum the total on random dice rolls.`
 3. `View this code at https://nostarch.com/big-book-small-python-projects`
 4. `Tags: large, artistic, game, math"""`
 - 5.
 6. `import random, time`
 - 7.
 8. `# Set up the constants:`
 9. `DICE_WIDTH = 9`
 10. `DICE_HEIGHT = 5`
 11. `CANVAS_WIDTH = 79`